
“Data Mining - Uncovering Consumer Purchase Patterns Using Market Basket Analysis and Frequent Pattern Mining”

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Abstract

The objective of this project is to find frequent patterns and use Market Basket Analysis to create association rules. By using Apriori and FP-Growth Algorithms and by leveraging concepts from Introduction to Data Mining, the project will examine the differences in results between each model and their performance with the dataset. Recommendations will be made based on the association rules found in the dataset.

1.0 Introduction and Problem Definition

The online retail industry has witnessed significant growth over the past decade, and understanding customer behaviour has become essential for improving business outcomes. This project focuses on analysing a real-world dataset from an online retail store to uncover insights into purchasing behaviour, identify customer segments, and provide recommendations that can guide strategic decisions. The primary goal is to apply the Apriori Algorithm and FP-Growth algorithm, we aim to identify frequent itemsets and generate association rules based on key metrics such as support, confidence, lift, and Kulczynski. In addition, build machine learning models that classify, and cluster customers based on their shopping behaviour, predict outcomes, and determine the most influential features driving customer decisions.

2.0 Data Collection

The dataset used in this analysis was sourced from the UCI Machine Learning Repository and contains transactions from a UK-based online retailer between 2010 and 2011. This is a large dataset with over 500,000 rows, which each represent a product and 8 total columns. The columns include an Invoice Number, Stock Code, Description, Quantity, Invoice Date, Unit Price, Customer ID, and Country.

3.0 Data Preprocessing

The initial step involved cleaning the data by removing null entries, negative or zero values for quantity and price, and converting date-time formats appropriately. Label encoding was applied to the 'Country' column, and normalization was done on numerical features such as 'Quantity' and 'UnitPrice'. Additional features were engineered, such as total spending per transaction and number of items per invoice. In order to use the Apriori Algorithm, which can be computationally expensive to run, we made sure to randomly sample invoices so we would be able to run this algorithm. Our dataset has more than 18,000 unique invoices, so we decided to use 3,000 randomly sampled invoices to run the Apriori and FP-Growth Algorithm on. These preprocessing steps ensured the dataset was well-structured and suitable for model training.

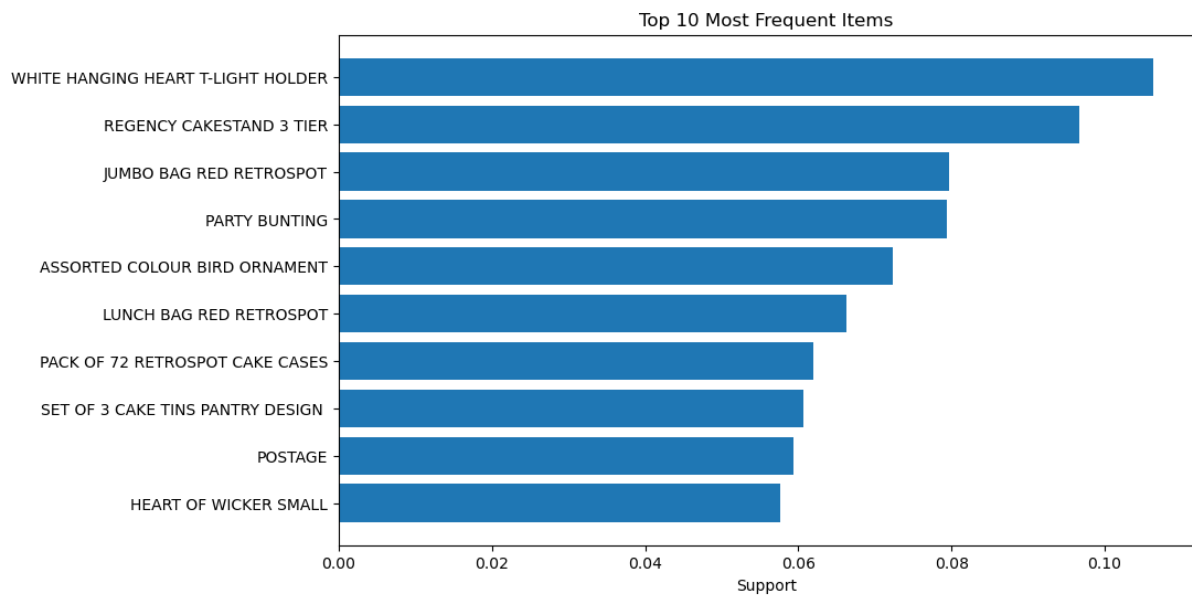
4.0 Problem Statement

In retail, it is very important to understand customer purchasing behaviour. Having a better understanding of this can help drive sales by better product placement and making the overall customer experience better. It can be challenging to know which products are often bought together and what relationships certain products have with each other. Luckily, by using data mining concepts and techniques, these patterns and relationships can be found and leveraged to improve business outcomes. The objective of this project is to use market basket analysis to discover frequent itemsets and generate association rules to support product bundling, cross-selling strategies, and personalized recommendations. By using Apriori and FP-Growth, along with a deeper dive in different evaluation metrics, we aim to find patterns in this particular dataset to give recommendations for better business outcomes.

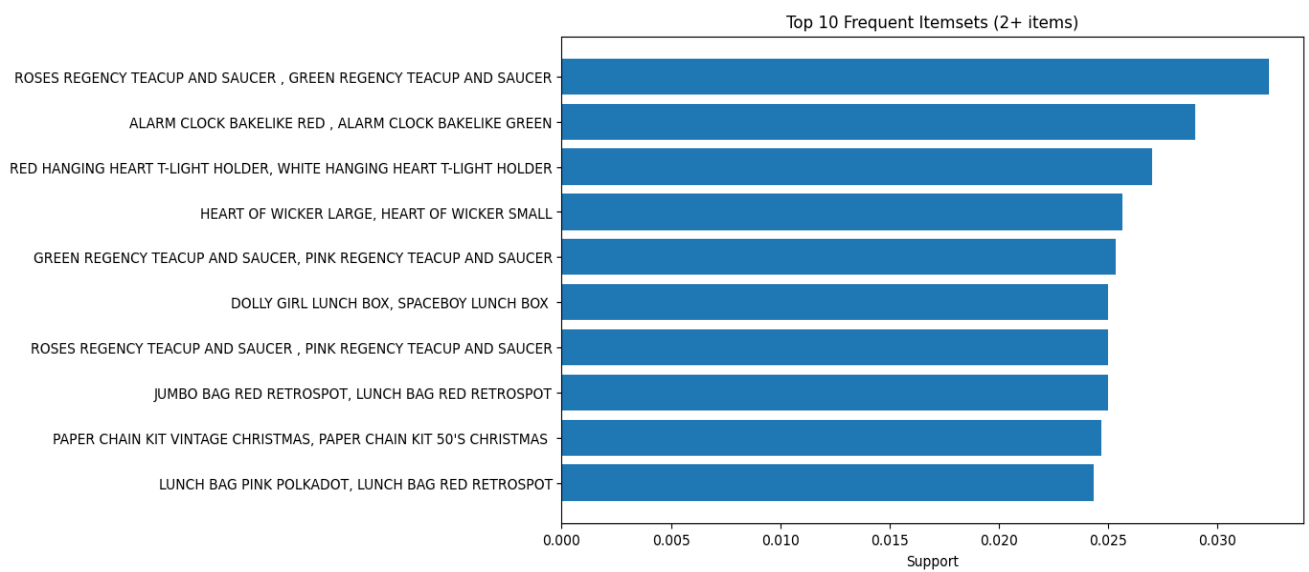
5.0 Implementation and Evaluation

The first model we used is the Apriori Algorithm. The Apriori Algorithm identifies frequent itemsets by generating candidate combinations and filtering them based on minimum support. The FP-Growth Algorithm works by compressing the dataset in a tree structure and then the algorithm mines the tree for recurring patterns. This method works more efficiently this way than the Apriori Algorithm because it doesn't generate candidates. Once the itemsets are found, they are measured using different metrics. One metric is called 'support', support measures how frequently a rule occurs in the dataset. 'Confidence' measures the conditional probability of B given A. 'Lift' measures how much more often A and B occur together than if they were independent. We decided to look at four more metrics which include Leverage, Conviction, Kulczynski, and Jaccard. Leverage measures the difference between the observed frequency of an itemset and the expected frequency if the items were independent. If this value is positive, then that means that the items occur together more often than we would expect by chance. Conviction compares the expected frequency that an antecedent occurs without its consequent, assuming independence, to its actual frequency. The higher this value is, the higher the implication is that if the antecedent was bought then the consequent will also be bought. The Kulczynski metric calculates the average of the confidence of both $A \rightarrow B$ and $B \rightarrow A$, making it a symmetric measure of association. The Jaccard index measures the proportion of transactions where both items occur together relative to transactions where occurs. This metric focuses on the overlap of the two items, and it shows how strongly the items co-occur. We this selection of metrics to focus on, it will show a full picture of the association rules. This will allow us to dive deep into what is truly happening and we can understand the generated rules better.

First, we applied the Apriori Algorithm with a minimum support of 0.02 and generated itemsets and association rules based. As mentioned before, this algorithm is computationally very demanding which is why we opted to randomly sample about 20% of the dataset's transactions to generate itemsets and rules with. Below are the top 10 most frequently found items in the dataset.



We also looked at frequent itemsets with 2 items because this would give us an idea of items that are frequently bought together.



We then started filtering the rules by confidence and lift. The table below shows the association rules where confidence is at least 0.8

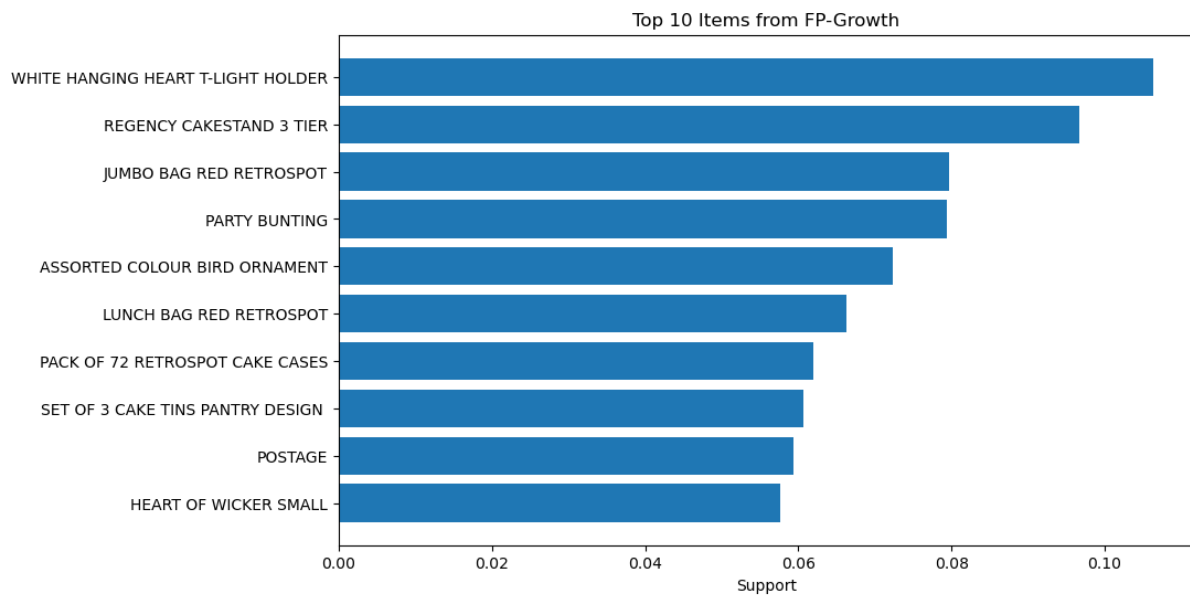
antecedents	consequents	support	confidence	lift
ROSES REGENCY TEACUP, PINK REGENCY TEACUP	GREEN REGENCY TEACUP AND SAUCER	0.022667	0.906667	22.857143
GREEN REGENCY TEACUP, PINK REGENCY TEACUP	ROSES REGENCY TEACUP AND SAUCER	0.022667	0.894737	19.592778
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.025333	0.844444	21.288515
PINK REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.025000	0.833333	18.248175
GREEN REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.032333	0.815126	17.849476

The table below shows all the association rules where Lift is the highest out of all rules.

antecedents	consequents	support	confidence	lift
PINK REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP, GREEN REGENCY TEACUP	0.022667	0.755556	23.367698

ROSES REGENCY TEACUP, GREEN REGENCY TEACUP	PINK REGENCY TEACUP AND SAUCER	0.022667	0.701031	23.367698
ROSES REGENCY TEACUP, PINK REGENCY TEACUP	GREEN REGENCY TEACUP AND SAUCER	0.022667	0.906667	22.857143
GREEN REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP, PINK REGENCY TEACUP	0.022667	0.571429	22.857143
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.025333	0.638655	21.288515
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.025333	0.844444	21.288515

Next, we ran the FP-Growth model to compare our association rules with. We used the same sample and same minimum support. The first thing that was noteworthy, was how much more efficiently the FP-Growth model ran. It was faster and it ultimately gave the same results, because we found the exact same number of frequent itemsets. The table below shows the 10 most frequent items using the FP-Growth Algorithm.



The top rules are also the same as the Apriori table where we filtered by Lift. This shows consistent performance between both methods, and it shows are strong these rules are.

Now, we wanted to look at these same rules and see how they perform with the abovementioned more advanced metrics. The table below shows those same six rules, but then with Leverage, Conviction, Kulczynski and Jaccard.

antecedents	consequents	Leverage	Conviction	Kulczynski	Jaccard
PINK REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP, GREEN REGENCY TEACUP	0.021697	3.958636	0.728293	0.571429
ROSES REGENCY TEACUP, GREEN REGENCY TEACUP	PINK REGENCY TEACUP AND SAUCER	0.021697	3.244483	0.728293	0.571429
ROSES REGENCY TEACUP, PINK	GREEN REGENCY TEACUP AND SAUCER	0.021675	10.289286	0.739048	0.539683

REGENCY TEACUP						
GREEN REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP, PINK REGENCY TEACUP	0.021675	2.275000	0.739048	0.539683	
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.024143	2.684419	0.741550	0.571429	
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.024143	6.173571	0.741550	0.571429	

All the item pairs show positive leverage values, indicating that these combinations occur together more often than expected by random chance. Conviction values vary since some pairs have a very high conviction of, for example, 10.29. While others have a lower conviction of 2.27. This would suggest that the consequent is a bit more common. Kulczynski scores are consistently high, suggesting a balanced, strong association in both directions. The Jaccard scores show a somewhat high degree of overlap between customers buying these combinations. Overall, these rules are very strong, and these specific products show a strong purchasing pattern

6.0 Findings and Recommendation

By using the Apriori and FP-Growth Algorithms we were able to find purchasing patterns in the retail dataset. These patterns are consistent across both methods which shows the robustness of the itemsets and associations rules found in the data. We analysed the rules using different metrics which helped us get a clear understanding of the rules. The metrics included support, confidence, lift, and more advanced metrics like leverage, conviction, Kulczyinski, and Jaccard.

Our key findings are that the Teacup products are the strongest rules and the different items in the Teacup series are often bought together. All metrics score high for these rules which

confirm the strength of the rules and relationship between the products. With extremely high lift of over 20, high confidence over 80%, high Kulczynski scores of around 0.73, positive leverage values and conviction values of up to 10.29 for some we conclude that, usually, when someone buys a specific colour teacup set, they are likely to buy another coloured set.

This finding is consistent with other products, such as lunchboxes and alarm clocks, that have different colour variations, as shown in one of the graphs above. This is also the case with some of the seasonal products that are in the dataset such as the Christmas décor products.

Some recommendations that we made based upon the results of our research is to bundle items that are naturally bought together, especially teacup sets, lunch boxes, and other products that have varying colours. These bundles could have promotional actions like a “Buy 2 get 1 Free” where customers can buy, for example, the rose, pink, and green teacup sets. Next, business can use seasonality and product design themes to craft curated bundles and recommend related items. Another recommendation is to implement cross-sell strategies by recommending complementary product designs on product pages or at checkout when a customer adds one product to their cart. So, in this case, when someone adds one of the teacup sets, we can promote and recommend another teacup set. Thirdly, these products should be physically grouped or digitally displayed together to encourage pattern behaviour. Since we know that the rules are bi-directional, discovered through Kulczynski and Jaccard scores, there is a greater chance that multiple products like the teacups are bought at the same time. Lastly, a business should maintain sufficient stock levels for these popular products to prevent lack of product, especially around peak seasons or promotional periods.

7.0 Conclusion

In this project we applied Market Basket Analysis methods to identify purchasing patterns in a retail dataset. We used both Apriori and FP-Growth Algorithms to find frequent itemsets and association rules to make insightful recommendations for businesses. The rules were analysed by using multiple different metrics to provide a clear picture of the association rules. The consistency across these metrics shows the potential of different strategies one can take to capitalize on the purchasing behaviour of customers. Bundling, cross-selling, and inventory management are the main recommendations based on the results of the research. In conclusion, this project shows the strength of potential insights gained from association rule mining, which can be helpful when making decisions in a business.

8.0 Reference

<https://archive.ics.uci.edu/dataset/352/online+retail>